```
#include <stdio.h>
int main() {
    int n, i, time = 0, completed = 0;
    int pid[20], at[20], bt[20], pr[20];
    int rt[20], ct[20], tat[20], wt[20];
    int highest, pos;
    float avg_wt = 0, avg_tat = 0;
    printf("1WA24CS218\n");
    printf("Enter number of processes: ");
    scanf("%d", &n);
    for(i = 0; i < n; i++) {
        pid[i] = i + 1;
        printf("P%d Arrival Time: ", i+1);
        scanf("%d", &at[i]);
        printf("P%d Burst Time: ", i+1);
        scanf("%d", &bt[i]);
        printf("P%d Priority: ", i+1);
        scanf("%d", &pr[i]);
        rt[i] = bt[i];
    }
    while(completed < n) {
        highest = 999;
        pos = -1;
        for(i = 0; i < n; i++) {
            if(at[i] <= time && rt[i] > 0 && pr[i] < highest) {
                highest = pr[i];
                pos = i;
            }
        }
        if(pos == -1) {
```

```
            time++;
        }
        else {
            rt[pos]--;
            time++;
            if(rt[pos] == 0) {
                completed++;
                ct[pos] = time;
                tat[pos] = ct[pos] - at[pos];
                wt[pos] = tat[pos] - bt[pos];
                avg_wt += wt[pos];
                avg_tat += tat[pos];
            }
        }
    }
    printf("\nProcess\tAT\tBT\tPR\tCT\tTAT\tWT\n");
    for(i = 0; i < n; i++) {
        printf("P%d\t%d\t%d\t%d\t%d\t%d\t%d\n",
               pid[i], at[i], bt[i], pr[i], ct[i], tat[i], wt[i]);
    }
    avg_wt /= n;
    avg_tat /= n;
    printf("\nAverage Waiting Time = %.2f", avg_wt);
    printf("\nAverage Turnaround Time = %.2f\n", avg_tat);
    return 0;
}
```

```
Enter number of processes: 5
P1 Arrival Time: 0
P1 Burst Time: 3
P1 Priority: 3
P2 Arrival Time: 1
P2 Burst Time: 4
P2 Priority: 2
P3 Arrival Time: 2
P3 Burst Time: 6
P3 Priority: 4
P4 Arrival Time: 3
P4 Burst Time: 4
P4 Priority: 6
P5 Arrival Time: 4
P5 Burst Time: 2
P5 Priority: 10

Process AT      BT      PR      CT      TAT     WT
P1      0       3       3       7       7       4
P2      1       4       2       5       4       0
P3      2       6       4       13      11      5
P4      3       4       6       17      14      10
P5      4       2       10      19      15      13

Average Waiting Time = 6.40
Average Turnaround Time = 10.20

Process returned 0 (0x0)   execution time : 34.233 s
Press any key to continue.
```